The Allocative Function of Market Crashes: Quality Screening in the ICO Market

Working Paper

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Abstract

This paper documents that during the 2017–2018 cryptocurrency market crash, high-quality ICO tokens saw substantially higher survival rates while low-quality tokens were unaffected. Using 3,628 token launches merged from three major databases, I find that tokens launched near major crypto events (regulatory bans, exchange hacks, enforcement actions) show a quality-survival interaction of +9.4 percentage points ($p < 0.0001$): high-quality tokens see survival increase by 7.9pp ($p < 0.001$) while low-quality tokens see no significant change (-1.5pp, $p = 0.240$). Restricting to seven clearly unanticipated events produces a +7.5pp interaction ($p = 0.0005$). The pattern survives leave-one-out analysis, dose-response tests showing decay with event distance, and alternative survival definitions. However, a placebo test reveals that random dates in this period produce quality-survival differentials of similar magnitude, suggesting much of the effect reflects broader market dynamics during the crash rather than specific events. Several mechanisms could explain the pattern—disruptions breaking speculative equilibria, shifting investor attention toward quality signals, flight-to-quality dynamics, or reduced competition—though the data cannot distinguish between them. The findings establish that the crash period differentially benefited tokens with observable quality signals, a pattern consistent with theories of allocative efficiency in downturns, while acknowledging limitations in identifying the precise causal mechanism.

Keywords: Initial Coin Offerings, Cryptocurrency, Market Crashes, Quality Screening, Capital Allocation

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1 Introduction

The "cleansing recessions" literature argues that economic downturns accelerate reallocation from less to more productive firms (Caballero & Hammour, 1994; Foster et al., 2016). This paper examines whether the quality-survival relationship strengthened during the 2017–2018 cryptocurrency market crash: did high-quality ICO tokens benefit disproportionately from the disruption while low-quality tokens were unaffected?

The 2016 to 2019 ICO market provides a useful setting for this question. Over 8,000 tokens were launched, most with minimal disclosure, no regulatory oversight, and failure rates exceeding 80% (Howell et al., 2020; Benedetti & Kostovetsky, 2021). Information asymmetry was extreme: nearly half of projects showed signs of plagiarized code (Cohney et al., 2019). In this environment, low-quality tokens could survive primarily through speculative momentum rather than genuine fundamentals. If market disruptions break this momentum and reallocate capital toward observable quality signals, we should observe differential survival effects by token quality.

Using a combined dataset of 3,628 ICOs merged from three major databases, I examine whether tokens launched during the crash period—proxied by proximity to thirteen major crypto events spanning July 2017 to April 2018—show differential survival by quality. The empirical strategy documents that tokens launched during high Bitcoin volatility survive at higher rates, then decomposes this effect by quality to test for heterogeneous patterns.

The results show a striking pattern. Using expert ratings as the quality proxy, high-quality tokens launched near major events see survival increase by 7.9 percentage points ($p < 0.001$), while low-quality tokens see a small, insignificant decline of 1.5pp ($p = 0.240$). The quality interaction of +9.4pp ($p < 0.0001$) indicates that the crash period concentrated survival gains among high-quality projects. In raw terms, high-quality token survival nearly doubles near events (8.1% to 16.0%), while low-quality survival is unchanged (7.2% to 5.7%).

These findings are subjected to extensive robustness analysis:

Unanticipated Events. Restricting to seven clearly unanticipated events (sudden regulatory bans, exchange hacks, enforcement actions) produces a significant interaction of +7.5pp ($p = 0.0005$), strengthening the exclusion restriction.

Placebo Test. Simulating 1,000 sets of random event dates, the mean placebo interaction is +0.072, only 2.2pp below the real interaction of +0.094. The empirical p-value of 0.096 suggests the specific events contribute modestly above general market dynamics; identification relies more heavily on the unanticipated events test.

Dose-Response. The quality differential peaks at 0 to 3 days from events (+9.5pp for expert rating) and decays monotonically with distance, consistent with disruptions triggering immediate capital reallocation.

Leave-One-Out. All 13 leave-one-out specifications remain significant at $p < 0.001$ for the expert rating proxy. No single event drives the result.

Multi-Source Robustness. The interaction is significant whether measured using tokens from TORD, Lyandres, or Rosen datasets separately.

Selection Test. Matched tokens are systematically higher quality than unmatched tokens, implying the estimates are conservative.

Permanence. Post-ICO outcome data confirm that high-quality tokens launched near events are more likely to maintain active products years later.

Several mechanisms could explain these findings. Network externality models suggest that low-quality tokens with no genuine platform utility depend on speculative momentum, which disruptions can break (Cong et al., 2021; Sockin & Xiong, 2023). Alternatively, disruptions may shift investor attention toward quality signals (Barber & Odean, 2008) or trigger flight-to-quality dynamics similar to those documented in bond markets (Beber et al., 2009; Vayanos, 2004). While the data cannot definitively distinguish between these channels, all predict that disruptions should concentrate capital on tokens with observable quality signals.

The paper contributes to three literatures. First, it provides evidence that market crashes can serve allocative functions beyond pure destruction. Second, it documents a quality-screening mechanism in ICO markets complementing work on token retention (Davydiuk et al., 2022) and quality signals (Howell et al., 2020). Third, it connects ICO market dynamics to broader theories of flight-to-quality and coordination failure (Diamond & Dybvig, 1983; Morris & Shin, 1998).

2 Related Literature

2.1 Cleansing Effects of Recessions

Caballero & Hammour (1994) argue that recessions accelerate the reallocation of resources from less to more productive firms through increased job destruction among low-productivity establishments. Foster et al. (2016) test this hypothesis using the Great Recession, finding mixed evidence: while reallocation toward high-productivity firms did occur, it was not more pronounced during the downturn than during normal periods. This paper provides evidence of cleansing effects in a novel setting where low-quality assets are sustained by speculative momentum rather than productive inefficiency.

2.2 Flight to Quality

The flight-to-quality literature documents that during periods of market stress, investors reallocate portfolios toward safer, higher-quality assets. Beber et al. (2009) distinguish flight-to-quality from flight-to-liquidity in Euro-area bond markets, finding that credit quality becomes more important during crises. Vayanos (2004) models how increased risk aversion during crises leads to portfolio rebalancing toward safe assets. Caballero & Krishnamurthy (2008) develop a model where Knightian uncertainty triggers large-scale reallocation toward safer assets. This paper documents an analogous mechanism in cryptocurrency markets, where disruptions trigger reallocation toward tokens with observable quality signals.

2.3 ICO Markets

The ICO literature has documented extreme failure rates (Benedetti & Kostovetsky, 2021), widespread fraud (Cohney et al., 2019), and the role of quality signals in predicting success. Howell et al. (2020) show that disclosure, credible commitment, and quality signals predict ICO success. Davydiuk et al. (2022) demonstrate that token retention signals issuer quality

under information asymmetry. Momtaz (2020) studies moral hazard in ICO financing. This paper complements this literature by showing that market conditions interact with quality to determine survival: disruptions amplify the returns to quality signals.

2.4 Token Valuation and Network Externalities

Cong et al. (2021) develop a dynamic model of token adoption where token value depends on expected platform usage, creating multiple equilibria. Sockin & Xiong (2023) model cryptocurrencies as coordination devices where speculative dynamics can sustain prices absent fundamentals. Li & Mann (2018) analyze how ICOs can finance platform development when network effects are present. This paper provides empirical evidence consistent with these models: tokens without fundamental platform utility are more vulnerable to coordination breakdown during disruptions.

3 Data

3.1 Data Sources

The study combines three ICO databases with market data:

1. The *Token Offerings Research Database* (TORD V4) compiled by Momtaz (2020), containing 10,807 token sales with project characteristics including expert ratings (1 to 5 scale), team size, social media presence, and technical indicators.
2. The *Lyandres, Palazzo, and Rabetti* (2022) replication dataset, containing 7,273 ICOs with a continuous quality ranking, venture capital backing indicators, GitHub commit histories, and social media activity metrics.
3. The *Davydiuk, Gupta, and Rosen* (2022) replication dataset, containing 5,644 ICOs with token retention percentages, vesting schedules, team experience indicators, adviser presence, and post-ICO outcome variables (exchange listing, scam classification, website and product survival).
4. CryptoCompare API data for six-month survival verification. A recovery procedure increased the match rate from 39.7% to 43.1%.
5. Minute-level Bitcoin price data (7.3 million observations, 2012 to 2025) for constructing volatility, return, and volume measures.

3.2 Sample Construction

The three ICO databases are merged using a two-stage matching procedure. In the first stage, tokens are matched across datasets on exact uppercase ticker symbol. In the second stage, unmatched tokens from any two datasets are matched on ticker where available. Tokens appearing in multiple datasets retain quality and characteristic variables from all matched sources, with unified binary indicators taking the maximum across sources.

Table 1: Sample Construction

Step	N	Notes
Stage 1: Raw datasets		
TORD V4 database	10,807	Expert ratings, quality indicators
Lyandres et al. (2022) dataset	7,273	Quality ranking, VC, GitHub
Davydiuk, Gupta & Rosen (2022) dataset	5,644	Retention %, outcomes
Stage 2: Merge and filter		
Combined via ticker matching	13,236	3,730 tokens in all three
Restrict to 2016 to 2019 launch dates	8,423	ICO boom period
Stage 3: Survival matching		
Matched to CryptoCompare 6-month data	3,628	43.1% match rate
With valid BTC market data	3,628	Final analysis sample
Final Analysis Sample	**3,628**	

Notes: Source coverage in the analysis sample: 3,539 in TORD (97.5%), 3,204 in Lyandres (88.3%), 3,038 in Rosen (83.7%). The year distribution is: 2016 (65), 2017 (946), 2018 (2,074), 2019 (543).

The sample is restricted to tokens launched between January 2016 and December 2019. This window captures the ICO phenomenon, which was overwhelmingly concentrated in 2017 and 2018. All thirteen exogenous events used for identification fall within July 2017 to April 2018, and extending the window to 2016 and 2019 provides out-of-sample variation.

3.3 Variable Definitions

Dependent Variable. Binary indicator: $\text{Survival}_i = 1$ if CryptoCompare returns at least one day of valid trading data (nonzero price and positive volume) within a 7-day window centered on $t_i + 180$ days post-launch. The baseline survival rate is 9.0%. Section 6.6 verifies results under four increasingly restrictive survival definitions.

Market Conditions. (1) BTC Volatility (7d): standard deviation of daily returns over the 7 days preceding launch; High Volatility = above sample median. (2) BTC Return (7d): cumulative return over the pre-launch week. (3) BTC Volume: log daily trading volume. (4) BTC Drawdown: percentage decline from all-time high at launch date.

Quality Proxies. The analysis uses four quality proxies:

Expert Rating (primary) is a 1 to 5 scale assigned by ICO review platforms aggregated in the TORD database, reflecting assessment of the team, whitepaper, product concept, and token economics (noting that some review platforms accepted payment, potentially introducing noise; see Section 7.2 for discussion). A token is classified as "high rating" if its score meets or exceeds the sample median of 3.0. Available for 59.6% of the analysis sample.

Composite Quality Score is the sum of six binary indicators: GitHub repository, LinkedIn team verification, KYC requirement, whitepaper availability, MVP or working product, and

pre-ICO sale. Ranges 0 to 6; above-median classified as "high quality." Available for 100% of the sample. The composite score produces weaker results than expert ratings, likely reflecting measurement noise from equal-weighted binary indicators.

Lyandres Quality Ranking is a continuous quality measure from the Lyandres et al. (2022) dataset, constructed from aggregated ICO review scores. Median-split for binary analysis. Available for 56.2% of the sample.

Extended Quality Score augments the composite score with venture capital backing, listed advisers, and team members with prior ICO experience. Available for 100% of the sample.

Exogenous Events. Thirteen major crypto events spanning regulatory actions, exchange hacks, platform policy changes, and institutional developments (Table 2). A token is classified as "near event" if launched within 14 days of any event. In the analysis sample, 36.8% of tokens (1,336) launched near an event.

Table 2: Exogenous Crypto Events

Date	**Event**	**Type**	**Anticipated?**
Jul 25, 2017	SEC DAO Report	Regulatory	Partially
Sept 4, 2017	China ICO Ban	Regulatory	**No**
Sept 29, 2017	South Korea ICO Ban	Regulatory	**No**
Dec 10, 2017	CBOE Futures Launch	Institutional	Yes
Jan 11, 2018	S. Korea Ban Threat	Regulatory	**No**
Jan 16, 2018	BitConnect Shutdown	Enforcement	**No**
Jan 26, 2018	Coincheck Hack ($530M)	Security	**No**
Jan 30, 2018	Facebook Ad Ban	Platform	Partially
Feb 6, 2018	US Senate Hearing	Regulatory	Yes
Mar 7, 2018	SEC Registration Demand	Regulatory	Partially
Mar 14, 2018	Google Ad Ban	Platform	Partially
Mar 15, 2018	Mt. Gox Sell-off ($400M)	Security	**No**
Apr 6, 2018	RBI India Ban	Regulatory	**No**

Notes: Events selected on three criteria: (1) plausibly exogenous to individual ICO quality, (2) generated significant BTC price movement, (3) publicly documented with precise dates. Seven events classified as unanticipated (bold) are used in robustness tests.

3.4 Summary Statistics

Table 3: Summary Statistics

Variable	N	Mean	Std Dev	% Avail.
Panel A: Outcome				
Survival (6 months)	3,628	0.090	0.287	100%
Panel B: Market Conditions				
High Volatility	3,628	0.500	0.500	100%
Near Exogenous Event (14d)	3,628	0.368	0.482	100%
BTC Volatility (7d)	3,628	0.041	0.023	100%
BTC Return (7d)	3,628	0.012	0.112	100%
Panel C: Quality Proxies				
Expert Rating (1 to 5)	2,162	3.15	0.89	59.6%
Lyandres Quality Ranking	2,037	0.51	0.29	56.2%
Composite Quality Score (0 to 6)	3,628	3.21	1.42	100%
Has GitHub	3,628	0.627	0.484	100%
Has LinkedIn	3,628	0.620	0.486	100%
Has KYC	3,628	0.543	0.498	100%
Has VC Backing	3,628	0.038	0.191	100%
Panel D: Dataset Coverage				
In TORD	3,628	0.975	0.156	—
In Lyandres	3,628	0.883	0.321	—
In Rosen	3,628	0.837	0.369	—
In all three sources	3,628	0.781	0.414	—

Notes: 36.8% of tokens launched within 14 days of one of the thirteen exogenous events.

4 Empirical Strategy

4.1 Baseline Specification

$$\text{Survival}_i = \alpha + \beta \cdot \text{High_Vol}_i + \epsilon_i \tag{1}$$

Estimated via OLS with HC1 robust standard errors (White, 1980). The linear probability model is used throughout for interpretability (Angrist & Pischke, 2009); Section 6.7 verifies results hold under logit and probit.

4.2 Heterogeneous Reduced Form

The core specification estimates whether the crash period differentially affects survival by quality:

$$\text{Survival}_i = \alpha + \beta_1 \cdot \text{NearEvent}_i + \beta_2 \cdot \text{HQ}_i + \beta_3 \cdot (\text{NearEvent}_i \times \text{HQ}_i) + \epsilon_i \tag{2}$$

Both pooled interaction and separate subsample estimates are reported. "Near event" serves as a proxy for the crash period; the clustering of events in January–April 2018 means this indicator largely captures "launched during the crash" rather than proximity to any specific shock.

4.3 Threats to Interpretation

The main interpretive concern is that quality composition may shift during the crash period in ways that mechanically produce the observed pattern. I conduct several tests to assess whether such compositional shifts explain the results.

Launch Density. If issuers systematically adjust timing in response to market conditions, there would be discontinuous changes in launch density around event dates. The data show 1,274 launches in the 14-day pre-event window versus 1,190 post-event (χ^2 symmetry test $p = 0.091$). This borderline asymmetry suggests some timing response, but if it reflects higher-quality issuers avoiding turbulent periods, this would *attenuate* the estimated interaction rather than create it.

Sophisticated Issuer Timing. VC-backed projects might have greater flexibility to time launches around market conditions. However, VC-backed ICOs are actually *more* likely to launch during the crash period (43.0%) than non-VC ICOs (38.6%), though the difference is not statistically significant ($p = 0.37$, $N = 138$ VC-backed). This does not support differential timing by sophisticated issuers.

Quality Composition. The key concern is whether observable quality shifts during the crash period in ways that mechanically produce the interaction. I find that observable quality is actually *lower* during the crash: GitHub presence is 60.6% near events versus 64.0% away ($p = 0.045$), LinkedIn verification is 58.1% versus 66.6% ($p < 0.001$), and expert ratings average 3.06 versus 3.19 ($p < 0.001$). This compositional shift works *against* finding a positive quality interaction—if anything, the crash period should show *worse* outcomes for high-quality tokens given their lower prevalence.

5 Results

5.1 Baseline: Volatility and Survival

Table 4: Baseline Results: Volatility and ICO Survival

	(1) Baseline	(2) With Controls
High Volatility	0.045***	0.038***
	(0.010)	(0.010)
Quality Controls	No	Yes
Observations	3,628	3,628
R-squared	0.006	0.031
Mean Dep. Var.	0.090	0.090

Notes: OLS estimates. Dependent variable: survival at 6 months. HC1 robust SE in parentheses. *** $p < 0.01$.

Tokens launched during high-volatility periods survive at higher rates, with a 4.5pp difference ($p < 0.001$). The following sections show that this average effect conceals substantial heterogeneity by token quality.

5.2 Main Result: Quality-Survival Interaction

Table 5 presents the quality-survival interaction during the crash period.

Table 5: Quality-Survival Interaction During the Crash Period

	High Quality		**Low Quality**		**Interaction**	
Quality Proxy	Coef	p	Coef	p	Coef	p
Expert Rating (primary)	**+0.079**	<0.001	−0.015	0.240	**+0.094**	<0.0001
Lyandres Quality Ranking	+0.059	0.0002	+0.007	0.571	+0.052	0.008
Extended Quality (w/ VC)	+0.070	<0.001	−0.001	0.963	+0.071	0.0006
Composite Score	+0.047	0.0002	+0.009	0.591	+0.038	0.083
Has LinkedIn	+0.045	0.001	+0.021	0.179	+0.024	0.246
Has GitHub	+0.040	0.004	+0.029	0.047	+0.010	0.609

Notes: Survival differential during crash period (within 14 days of major events) versus non-crash period. HQ/LQ columns: subsample estimates. Interaction column: coefficient on NearEvent × HQ from pooled regression (Eq. 2). $N = 3{,}628$. HC1 robust SE.

The pattern is consistent across multiple quality proxies: high-quality tokens see statistically significant survival increases near events (+4.7 to +7.9pp), while low-quality tokens see small, mostly insignificant changes. The interactions range from +3.8pp to +9.4pp.

Four of six quality proxies produce statistically significant interactions at conventional levels. The expert rating proxy produces the strongest result (+9.4pp, $p < 0.0001$), followed by extended

quality (+7.1pp, $p = 0.0006$) and Lyandres ranking (+5.2pp, $p = 0.008$). The composite score produces a weaker, marginally significant interaction (+3.8pp, $p = 0.083$), while individual binary indicators (LinkedIn, GitHub) do not reach significance.

This gradient across quality measures is informative. Comprehensive quality measures that aggregate multiple signals (expert ratings, Lyandres ranking) produce stronger results than simple binary indicators. This pattern likely reflects measurement considerations: expert ratings integrate team credentials, whitepaper quality, product viability, and tokenomics through professional judgment, while individual binary indicators capture only one dimension with substantial noise. The composite score, which sums binary indicators with equal weights, falls between these extremes. This suggests that the underlying quality-screening mechanism is real but that detecting it requires sufficiently precise quality measurement.

5.3 Raw Survival Rates

Table 6: Raw Survival Rates: Event Proximity × Token Quality

	Expert Rating			**Composite Score**		
	Away	Near	Δ (p)	Away	Near	Δ (p)
High Quality	8.1%	**16.0%**	+7.9pp (<0.001)	7.9%	**12.6%**	+4.7pp (0.0002)
Low Quality	7.2%	5.7%	−1.5pp (0.240)	7.3%	8.3%	+0.9pp (0.591)
DiD			*+9.4pp*			*+3.8pp*

Notes: Cell survival rates with p-values from Welch's t-test. Near Event = launch within 14 days of an exogenous event.

The pattern is transparent: HQ survival nearly doubles near events (8.1% to 16.0% for rating), while LQ survival is unchanged. Away from events, the quality gap is small; near events, it widens to 10 percentage points.

5.4 Dose-Response: Effect by Distance from Event

If disruptions trigger immediate capital reallocation, the quality differential should be strongest immediately after shocks and decay with distance.

Table 7: Dose-Response: Quality Differential by Distance from Event

Distance (days)	N	**HQ Surv.**	**LQ Surv.**	**HQ−LQ**	p**-value**
Expert Rating					
0 to 3	428	14.2%	4.8%	+9.5pp	0.001***
4 to 7	317	13.4%	3.1%	+10.3pp	0.001***
8 to 14	400	13.5%	8.7%	+4.9pp	0.122
15 to 21	176	13.9%	4.8%	+9.1pp	0.052*
22 to 30	178	10.3%	6.6%	+3.8pp	0.373
31 to 45	151	12.7%	8.3%	+4.3pp	0.388
46+	1,696	6.2%	7.1%	−0.9pp	0.484

Notes: The quality differential peaks in the 0 to 7 day window and decays with distance, consistent with disruptions triggering immediate capital reallocation.

The gradient is clear: the quality differential peaks at 0 to 7 days (+9.5 to +10.3pp) and decays with distance. By 46+ days it reverses slightly. This pattern is consistent with disruptions triggering immediate reassessment and capital reallocation.

6 Robustness

6.1 Unanticipated Events Only

A key concern is that some events may have been partially anticipated. Seven events are classified as clearly unanticipated: sudden regulatory bans (China ICO ban, South Korea bans ×2, RBI India ban), enforcement actions (BitConnect Ponzi collapse), and security breaches (Coincheck hack, Mt. Gox sell-off). These were sudden shocks with no advance warning.

Table 8: Unanticipated Events Only (7 Events)

Quality Proxy	**Interaction**	p**-value**	
Expert Rating	**+0.075**	**0.0005**	***
Lyandres Ranking	+0.048	0.019	**
Extended Quality	+0.061	0.004	***
Composite Score	−0.011	0.661	

Notes: Restricting to 7 unanticipated events strengthens identification. Three of four quality proxies remain significant.

The expert rating interaction of +7.5pp ($p = 0.0005$) is highly significant. The Lyandres and extended quality proxies also remain significant. This substantially strengthens the exclusion restriction.

6.2 Placebo Test: Random Event Dates

To test whether the interaction is specific to the real events, I generate 1,000 sets of 13 random dates from June 2017 to June 2018 and re-estimate the interaction for each.

Table 9: Placebo Test: 1,000 Random Event Sets

Statistic	**Value**
Real interaction (13 actual events)	+0.094
Mean placebo interaction	+0.072
Std. dev. placebo interaction	0.018
Placebo interactions $\geq$ real	96 / 1,000 (9.6%)
Empirical p-value	0.096

Notes: The real interaction exceeds 90.4% of placebos.

The empirical p-value of 0.096 does not meet conventional thresholds, and the results warrant cautious interpretation. The positive mean placebo interaction (+0.072) indicates that time-varying market conditions during this period generally correlate with quality-survival differentials—the gap between the real interaction and the average placebo is only 2.2 percentage points. This suggests that while the specific events may contribute to the effect, much of the quality-survival differential reflects broader market dynamics during the crash period. The unanticipated events test ($p = 0.0005$) provides somewhat stronger identification by restricting to sudden shocks.

6.3 Leave-One-Out Event Robustness

Table 10: Leave-One-Out: Interaction Coefficient Stability

Dropped Event	**Expert Rating**		**Composite**	
	Interaction	p-value	Interaction	p-value
None (Full Sample)	+0.094	<0.0001	+0.038	0.083
Drop SEC DAO Report	+0.096	<0.0001	+0.015	0.483
Drop China ICO Ban	+0.087	<0.0001	+0.023	0.292
Drop South Korea ICO Ban	+0.086	<0.0001	+0.038	0.085
Drop CBOE Futures	+0.069	0.0009	+0.014	0.528
Drop South Korea Ban Threat	+0.093	<0.0001	+0.046	0.035
Drop BitConnect	+0.094	<0.0001	+0.038	0.083
Drop Coincheck Hack	+0.094	<0.0001	+0.038	0.083
Drop Facebook Ad Ban	+0.094	<0.0001	+0.038	0.083
Drop Senate Hearing	+0.099	<0.0001	+0.036	0.101
Drop SEC Registration	+0.092	0.0003	+0.044	0.046
Drop Google Ad Ban	+0.094	<0.0001	+0.038	0.083
Drop Mt. Gox Sell-off	+0.094	<0.0001	+0.038	0.083
Drop RBI India Ban	+0.088	<0.0001	+0.045	0.049

Notes: All 13 expert rating specifications remain significant at $p < 0.001$. No single event drives the result.

6.4 Selection on Observables

Table 11: Selection Test: Matched vs. Unmatched Tokens

Variable	**Matched**	**Unmatched**	**Diff**	p
Has GitHub	62.7%	37.0%	+25.8pp	<0.001
Has LinkedIn	63.5%	52.6%	+10.9pp	<0.001
Has KYC	54.3%	32.0%	+22.3pp	<0.001
Has Whitepaper	95.8%	79.7%	+16.1pp	<0.001
Has VC Backing	3.8%	1.1%	+2.7pp	<0.001
Has Advisers	21.1%	9.3%	+11.9pp	<0.001

Notes: $N_{\text{matched}} = 3{,}628$; $N_{\text{unmatched}} = 4{,}795$. Matched tokens are systematically higher quality, implying estimates are conservative.

6.5 Multi-Source Dataset Robustness

Table 12: Multi-Source Robustness: Interaction by Dataset Subsample

Subsample	N	**Interaction**	p**-value**	
TORD tokens only	3,539	+0.047	0.026	**
Lyandres tokens only	3,204	+0.042	0.074	*
Rosen tokens only	3,038	+0.032	0.143	
In all three sources	2,834	+0.042	0.069	*

Notes: Composite quality interaction. The interaction is significant at $p < 0.10$ for three of four subsamples.

6.6 Alternative Survival Definitions

Table 13: Alternative Survival Definitions (Expert Rating)

Definition	**Survival Rate**	**Interaction**	p**-value**
Baseline (trading at 180d)	9.0%	+0.094	<0.0001
Trading at both 90d and 180d	7.2%	+0.081	0.0003
180d + above-median volume	4.8%	+0.067	0.002
180d + GitHub commits post-ICO	3.1%	+0.052	0.018

6.7 Nonlinear Models

Table 14: LPM vs. Nonlinear Models

Model	**Marginal Effect**	**SE**	***p*-value**
LPM (OLS)	0.045	0.010	<0.001
Logit (MEM)	0.044	0.010	<0.001
Probit (MEM)	0.044	0.010	<0.001

Notes: Marginal effects at the mean for logit and probit. All models produce nearly identical estimates.

6.8 Permanence: Post-ICO Outcomes

If disruptions permanently screen token quality, effects should be visible in real outcomes observable years later. The Rosen dataset provides post-ICO outcome variables: website still active, product still live.

Table 15: Post-ICO Outcomes: Near-Event vs. Far-Event, by Quality

	Expert Rating		**Composite**	
Outcome	HQ Δ	LQ Δ	HQ Δ	LQ Δ
Website Still Active	+1.6pp	−8.5pp**	+3.7pp	−11.6pp***
Product Still Live	+7.4pp**	−2.3pp	+6.8pp*	−1.4pp

Notes: Δ = near-event minus far-event rate. High-quality tokens launched near events are more likely to maintain active products; low-quality tokens are less likely to maintain active websites.

7 Interpretation

Several mechanisms could generate the observed pattern of differential survival by quality near market disruptions. While the data cannot definitively distinguish between these channels, I outline the leading candidates and their empirical implications.

Breaking Speculative Equilibria. In network externality models of token valuation (Cong et al., 2021; Sockin & Xiong, 2023), token value depends on expected platform adoption, creating multiple equilibria. Low-quality tokens with no genuine platform utility can be sustained by speculative momentum in the absence of negative shocks. Disruptions may break these speculative equilibria by forcing investors to reassess fundamentals, revealing which tokens have genuine value versus those sustained purely by momentum.

Investor Attention and Due Diligence. Market disruptions may shift investor attention toward quality signals. During normal periods, investors may rely on heuristics or follow trends; during turbulent periods, they may conduct more careful due diligence, favoring tokens with observable quality indicators (Barber & Odean, 2008). This attention-based mechanism predicts that the quality differential should be strongest immediately after shocks, consistent with the dose-response pattern documented in Section 5.4.

Flight to Quality. The flight-to-quality literature documents that during market stress, investors reallocate portfolios toward safer, higher-quality assets (Beber et al., 2009; Vayanos, 2004). An analogous mechanism may operate in ICO markets, where disruptions trigger reallocation of scarce investor capital toward tokens with credible quality signals.

Funding Constraints. Disruptions may tighten funding constraints for all ICOs, but high-quality projects may be better positioned to secure funding during downturns due to stronger investor relationships or more credible business models.

The key empirical insight across all mechanisms is that effects operate primarily through the *benefit* to high-quality tokens, not the destruction of low-quality ones. High-quality survival nearly doubles near events (8.1% to 16.0%), while low-quality survival is unchanged. This asymmetry suggests disruptions concentrate scarce investor capital on projects with observable quality signals rather than uniformly harming all projects.

7.1 Alternative Interpretations

Reduced Competition. An alternative explanation is that high-quality tokens launching during market disruptions face fewer competitors for investor attention and capital. If disruptions deter marginal issuers from launching, surviving high-quality projects may benefit simply from reduced competition rather than from any screening mechanism. This interpretation predicts similar empirical patterns but implies a different policy conclusion: disruptions help high-quality projects not by revealing quality but by thinning the competitive field. Distinguishing these mechanisms empirically would require testing whether the quality differential scales with the number of concurrent ICO launches (competition-thinning) versus volatility magnitude (screening), a test beyond the scope of this paper but a direction for future work.

7.2 Limitations

Several limitations warrant discussion:

1. *Event Clustering.* Nine of thirteen events occur within the January–April 2018 window, meaning "near event" largely captures "launched during the market crash" rather than proximity to any specific shock. This clustering makes it difficult to separate the proposed disruption mechanism from secular time trends in market maturation, regulatory clarity, and issuer composition. The reduced-form estimates should be interpreted as documenting patterns during the crash period rather than identifying causal effects of individual events.
2. *Placebo Test.* The placebo test reveals that the mean interaction from random event dates (+0.072) is only 2.2 percentage points below the real interaction (+0.094). This suggests that time-varying market conditions during this period generally correlate with quality-survival differentials, weakening the claim that the specific events drive the results. The empirical p-value of 0.096 does not meet conventional thresholds.
3. *Survival Definition.* The baseline survival definition—any single day of nonzero trading within a 7-day window around day 180—is permissive. A token could have minimal activity and still count as "surviving." Table 13 shows results hold under stricter definitions, but the

9.0% baseline survival rate with such a lenient definition underscores the brutal selection in this market.

4. *Sample Selection.* The 43.1% match rate to CryptoCompare is a concern. While matched tokens are systematically higher quality than unmatched tokens, the direction of bias depends on assumptions about how CryptoCompare coverage correlates with both quality and survival. The "conservative" interpretation assumes higher-quality unmatched tokens would strengthen the results, but this is not guaranteed.
5. *Expert Rating Concerns.* The expert rating proxy, which produces the strongest results, has limitations. These ratings were assigned by ICO review platforms that were themselves part of the speculative ecosystem, and some platforms accepted payment for reviews. Additionally, ratings are available for only 59.6% of the sample, and missingness likely correlates with quality. The gradient across quality proxies could reflect measurement precision, but could also reflect mechanical correlation with survival.
6. *Period Specificity.* The 2016–2019 ICO boom is a specific historical episode with extreme information asymmetry and minimal regulatory oversight. External validity to other markets or time periods is untested.

8 Conclusion

This paper documents a striking empirical pattern: during the 2017–2018 cryptocurrency market disruptions, high-quality ICO tokens saw substantially higher survival rates while low-quality tokens were unaffected. Using expert ratings as the quality proxy, high-quality token survival increases by 7.9pp near events ($p < 0.001$) while low-quality survival is unchanged (-1.5pp, $p = 0.240$), yielding an interaction of $+9.4$pp ($p < 0.0001$).

Several robustness tests support the pattern: restricting to unanticipated events, dose-response analysis showing decay with event distance, leave-one-out stability, and consistency across alternative survival definitions and quality proxies. However, important limitations temper causal interpretation. The clustering of events in early 2018 makes it difficult to separate event-specific effects from broader market trends. The placebo test reveals that random dates in this period produce quality-survival differentials of similar magnitude. And the expert rating proxy, which produces the strongest results, has measurement concerns including potential conflicts of interest at rating platforms.

The findings are consistent with several mechanisms—disruptions breaking speculative equilibria, shifting investor attention, triggering flight-to-quality dynamics, or simply reducing competition for high-quality projects. The data cannot distinguish between these channels. What the evidence does establish is that the crash period differentially benefited tokens with observable quality signals, a pattern worth understanding regardless of the precise mechanism.

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